MZUZU UNIVERSITY

Department of Mathematics

EMT 1101 - Precalculus

End of Semester Examination

Date: December 7, 2015 **Time: 8:00 - 10:00 (3 hours)**

1. Check that the paper has **13 questions**.
2. Attempt all questions
3. Write answers in the space provided
4. Use proper notation
5. Simplify your answers as far as possible
6. **No Calculators**

Name _____

Reg Number , Programme: _____

Professor's / Lecturer's name: _____

1. Simplify the expression $\left(\frac{x^4y^{-8}z^2}{xy^2z^{-6}}\right)^{-2}$ (3)

Solution: $\left(\frac{x^4y^{-8}z^2}{xy^2z^{-6}}\right)^{-2} = \left(\frac{x^{4-1}z^{2+6}}{y^{8+2}}\right)^{-2}$
$= \left(\frac{x^3z^8}{y^{10}}\right)^{-2} = \frac{x^{-6}z^{-16}}{y^{-10}} = \frac{y^{10}}{x^6z^{16}}$

2. Rewrite the expression using rational exponents rather than radicals $\sqrt[3]{\sqrt{x}} + \sqrt{\sqrt[3]{x}}$ (3)

Solution: $\sqrt[3]{\sqrt{x}} + \sqrt{\sqrt[3]{x}} = \left(x^{\frac{1}{2}}\right)^{\frac{1}{3}} + \left(x^{\frac{1}{3}}\right)^{\frac{1}{2}} = x^{\frac{1}{6}} + x^{\frac{1}{6}} = 2x^{\frac{1}{6}}$

3. Carry out the indicated operations and simplify where possible $\frac{\left(a+\frac{1}{b}\right)^a\left(a-\frac{1}{b}\right)^b}{\left(b+\frac{1}{a}\right)^a\left(b-\frac{1}{a}\right)^b}$ (5)

Solution: $\frac{\left(a+\frac{1}{b}\right)^a\left(a-\frac{1}{b}\right)^b}{\left(b+\frac{1}{a}\right)^a\left(b-\frac{1}{a}\right)^b} = \frac{\left(\frac{ab+1}{b}\right)^a\left(\frac{ab-1}{b}\right)^b}{\left(\frac{ab+1}{a}\right)^a\left(\frac{ab-1}{a}\right)^b} = \frac{\frac{(ab+1)^a}{b^a}\frac{(ab-1)^b}{b^b}}{\frac{(ab+1)^a}{a^a}\frac{(ab-1)^b}{a^b}}$
Multiplying the numerator and denominator by $a^{a+b}b^{a+b}$ yields
$\frac{a^{a+b}(ab+1)^a(ab-1)^b}{b^{a+b}(ab+1)^a(ab-1)^b} = \frac{a^{a+b}}{b^{a+b}} = \left(\frac{a}{b}\right)^{a+b}$

4. Simplify $|x+1| + 4|x+3|$ given that $x < -3$ (2)

Solution: For $x < -3$ then $x+1$ and $x+3$ are both negative So we get $(x+1) - 4(x+3) = -x - 1 - 4x - 12 = -5x - 13$

5. (a) Solve the equation $2x^2 = 3 - 4x$ and leave answer in irrational form (4)

Solution: Re-arranging we have $2x^2 + 4x - 3 = 0$ and using quadratic formula or complete the square method We have $x = \frac{-b\pm\sqrt{(b^2-4ac}}{2a}$ where $a = 2, b = 4, c = -3$ which yields $x = \frac{-4\pm\sqrt{(4^2-4(2)(-3))}}{2(2)} = \frac{-4\pm\sqrt{16+24}}{4} = \frac{-2\pm\sqrt{10}}{2}$ so x is either $\frac{-2+\sqrt{10}}{2}$ or $\frac{-2-\sqrt{10}}{2}$ (4 marks)

(b) Show that triangle ABC with vertices $A(-7,4)$, $B(-3,10)$, $C(1,3)$ is isosceles triangle (4)

Solution: Use distance formula to find $AB = \sqrt{4^2+6^2} = \sqrt{52}$; $BC = \sqrt{4^2+(-7)^2} = \sqrt{65}$; $AC = \sqrt{8^2+(-1)^2} = \sqrt{65}$ Since sides $BC = AC$ then triangle ABC is isosceles

6. (a) Find the slope of the line passing through the two points $(3, 9)$ and $(3+h, (3+h)^2)$. (2)

Solution: Slope= $\frac{(3+h)^2-9}{3+h-3} = \frac{9+6h+h^2-9}{h} = 6+h$

(b) The center of a circle is the point $(3, 2)$. If the point $(-2, -10)$ lies on this circle, find the standard equation of the circle . (4)

Solution: Use distance formula to find the radius i.e $\sqrt{(-2-3)^2+(-10-2)^2} = \sqrt{169} = 13$. So the standard equation of circle is $(x-3)^2+(y-2)^2 = 13^2$

7. Find the value of the constant k for which $x^2+2(k+1)x+k^2 = 0$ has equal roots. (4)

Solution: $a = 1, b = 2(k+1), c = k^2$. For equal roots $b^2-4ac = 0$. So
$[2(k+1)]^2 - 4(1)(k^2) = 0$
$4(k^2+2k+1) - 4k^2 = 0$
$4k^2+8k+4-4k^2 = 0$
$8k = -4$
$k = -\frac{1}{2}$

8. Solve the inequality $\frac{3x}{5} - \frac{x-1}{3} < 1$ and express your answer in interval notation. (5)

Solution: $\frac{3x}{5} - \frac{x-1}{3} < 1$
$\frac{3x}{5} - \frac{x-1}{3} - 1 < 0$
$\frac{9x-5x+5-15}{15} < 0$
$\frac{4x+-10}{15} < 0$
$4x-10 < 0$
$x < \frac{5}{2}$
So the solution set is $\left(-\infty, \frac{5}{2}\right)$

9. (a) Let $h(x) = x^2-4x-c$. Find a nonzero value for c such that $h(c) = c$. (3)

Solution: $h(c) = c$ implies $c^2-4c-c = c$ from which we have $c^2-6c = 0$ $c(c-6) = 0$ which yields $c = 0$ or $c = 6$, hence the sought nonzero value is 6.

(b) The final grade in a course is a combination of a continuous assessment grade and the grade that the student scores in an end of semester exam. The minimum grade, as a percentage, that a student needs to score in an end of semester exam in order to obtain a pass in the course is given by the function (3)

$$g(x) = 75 - \frac{2x}{3}$$

where x is the continuous assessment grade as a percentage. If a student has a continuous assessment grade of 48%, what is the minimum grade that the student needs to score in the end of semester examination to obtain a final grade of pass?

Solution: The grade is $g(48) = 75 - \frac{2}{3} \times 48 = 75 - 2 \times 16 = 75 - 32 = 43\%$

10. Let $f(x) = \dfrac{2x-3}{x+4}$.

(a) Find the domain and range of the function $f(x)$ and express them using interval notation. (4)

Solution: The domain of $f(x)$ is the set of all real numbers excluding -4 i.e. the set $(-\infty, -4) \cup (-4, \infty)$.
To find the range let $y = \frac{2x-3}{x+4}$, making x the subject of formula we have obtain $x = \frac{4y+3}{2-y}$. Hence the range is the set $(-\infty, 2) \cup (2, \infty)$

(b) Find $f^{-1}(x)$. (3)

Solution: Let $y = \frac{2x-3}{x+4}$, interchanging x and y we have $x = \frac{2y-3}{y+4}$. Making y the subject of formula we have $y = \frac{4x-3}{x-2}$, hence $f^{-1}(x) = \frac{4x-3}{x-2}$

(c) Find the domain and range of the function $f^{-1}(x)$. What do you observe? (4)

Solution: The domain of f^{-1} is $(-\infty, 2) \cup (2, \infty)$
Now let $y = \frac{4x-3}{x-2}$, making x subject of formula we have $x = \frac{2y-3}{y-4}$, so the range of f^{-1} is $(-\infty, -4) \cup (-4, \infty)$. The observation is that the domain of f is the range of f^{-1} and the range of f is the domain of f^{-1}

11. (a) Graph the quadratic function (5)

$$y = -3x^2 + 12x.$$

(b) Use your graph to specify the vertex (turning point), axis of symmetry, maximum or minimum value and intercepts of $y = -3x^2 + 12x$. (5)

Solution: $y = -3x^2 + 12x = -3(x-2)^2 + 12$. Turning point: $(2, 12)$, axis of symmetry: $x = 2$, maximum value:$= 12$, x-intercepts: 0,4, y-intercept: 0

12. (a) Write the equation $\sqrt{2} = 2^{\frac{1}{2}}$ in logarithmic form (2)

    (b) Evaluate $\log_{16} \frac{1}{64}$ (3)

13. Find the domain of the function (6)

$$\log_{10} \frac{2x+3}{x-5}$$

Solution: Solve the inequality $\dfrac{2x+3}{x-5} > 0$. Key numbers are $x = -\frac{3}{2}$ and $x = 5$.

Interval	Test no.	$2x+3$	$x-5$	$\frac{2x+3}{x-5}$
$(-\infty, -\frac{3}{2})$	-2	neg.	neg.	pos.
$(-\frac{3}{2}, 5)$	0	pos.	neg.	neg.
$(5, \infty)$	6	pos.	pos.	pos.

Domain is $(-\infty, -\frac{3}{2}) \cup (5, \infty)$